Explainable Financial Alerts for Retail Investors: Integrating Statistical Anomaly Detection and News–Market Impact Correlation

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***Abstract*—Retail investors face a continuous flood of price movements and financial news but lack the interpretive tools available to institutions. We present InvestiGator, an explainable (XAI-first) financial-alert system for the US equity market that raises Telegram alerts from two independent triggers (abrupt price movements, detected as statistical anomalies, and incoming news) and attaches to every alert a fully traceable explanation. The core is a news–market correlation engine that retrieves analogous historical news with sentence embeddings and measures the impact those precedents had, in the manner of an event study and without lookahead. As an Artificial Intelligence Engineering contribution, the work integrates, applies and critically evaluates existing components in a functional, reproducible system. On real data, semantic retrieval reaches a cross-ticker precision@5 of 0.514 ± 0.015 against 0.240 for random and 0.346 for a lexical baseline, and a volatility-normalised detector fires far more consistently across instruments (firing-rate range 0.015) than a fixed threshold (0.344). A supervised materiality-triage model, trained on $79{,}753$ multi-year news–market examples, nearly quadruples within-budget alert precision (0.632 vs 0.163), although no model reading the headline text beat a volatility-only baseline, a finding we report as it stands. All results are reproducible from versioned scripts, with limitations stated explicitly.**

***Index Terms*—explainable AI, retail investing, anomaly detection, sentence embeddings, information retrieval, event study, financial NLP**

I. Introduction

Most Americans now own stock, and a growing share trade through commission-free mobile apps rather than through professional institutions [1]; the US equity market reached US$62.2 trillion at the end of 2024 [2]. Yet the non-professional, or *retail*, investor must contend with continuous price movements and a constant stream of news without the tools that banks and funds take for granted. Meanwhile, the AI now spreading through finance is largely opaque [3], which is a poor fit for someone who must act on an alert and live with the result.

This paper presents **InvestiGator**, an explainable financial-alert system that does *not* attempt to predict prices. Instead, for each abrupt move or relevant news item, it shows what it detected, how it reached that conclusion, the sources, and which past events resemble the current one, alongside what happened to prices afterwards. The contribution is one of Artificial Intelligence *Engineering*: not a new algorithm, but the disciplined integration, application and critical evaluation of existing components into a working, explainable and reproducible system, together with a documented methodology for relating news to market impact and a trained, calibrated materiality-triage model that ranks alerts without ever predicting price direction.

II. Related Work

Anomaly detection. Financial anomaly detection is dominated by unsupervised methods because labelled anomalies are scarce [4]. A rolling z-score is a transparent statistical detector; more expressive alternatives such as Isolation Forest [5] or deep detectors [6] trade interpretability for capacity. The need to normalise by recent volatility reflects volatility clustering, formalised by the ARCH/GARCH family [7], [8]; we use a non-parametric rolling estimate for transparency.

Explainability and trust. Surveys map the field and its transparent-versus-post-hoc divide [3]; a influential position argues for models that are interpretable by design in high-stakes settings [9], rather than post-hoc explanations of black boxes [10], [11]. Human-factors work frames the goal as *appropriate reliance* [12], and recent evidence warns that explanations can induce over-reliance [13]; we therefore explain with verifiable evidence rather than persuasive narrative.

Text representation and retrieval. Financial text analysis has moved from lexicons to learned representations; Sentence-BERT [14], built on the Transformer [15], yields directly comparable sentence vectors. Retrieval and its evaluation belong to information retrieval: the vector-space model [16], lexical ranking such as BM25 [17], and precision@k as the standard ranked-retrieval metric [18].

Event studies and case-based reasoning. The event-study design measures returns around an event [19], [20]; market efficiency [21] explains why we *measure* reactions rather than predict them. Using retrieved past cases as evidence is case-based reasoning [22]. The FNSPID dataset [23] time-aligns news with prices, enabling a news-to-impact knowledge base.

Learned triage. Deciding *which* news deserves an alert can be posed as supervised classification with event-study labels, without predicting direction. We use logistic regression, interpretable by design in the sense of [9], against gradient boosting

[24] as a capacity ceiling, with probabilities calibrated by a sigmoid fitted on validation data (Platt scaling) [25].

III. The InvestiGator System

InvestiGator is a pipeline of single-responsibility components linked by a common schema. Two data layers share that schema: a *historical* layer (a knowledge base of past news, each entry storing date, ticker, headline, sentence embedding and measured post-event impact) and a *live* layer (free price and news feeds). Two triggers converge on one explanation step before delivery via Telegram.

Market trigger. Let r_t be the log-return on day t and let μ_t, σ_t be the mean and standard deviation over a trailing window of w days ending strictly before t. The day is flagged when $|z_t| > k$ with $z_t = (r_t - \mu_t)/\sigma_t$. Every quantity behind the decision is exposed to the explanation engine, so the alert is faithful by construction.

News trigger. An incoming headline is embedded with Sentence-BERT and compared by cosine similarity against the knowledge base; the top-k precedents are returned with their observed impact at $+1$, $+3$ and $+5$ trading days, measured from the event-day close. The event day is the first trading day on or after the news date, and impact is accumulated strictly forward, which enforces a no-lookahead guarantee. This retrieve-and-reuse step is the case-based-reasoning core of the system.

Learned triage. On top of the two transparent rules, a supervised model estimates the probability that an abnormal move of unusual size, in either direction, follows a news item. Labels come from the system's own event-study code: an example is material when the magnitude of the ticker's return minus the market's over the three trading days after the event exceeds 2%. Features respect a strict timing convention (pre-event volatility and momentum, the event-day return, headline length, sector, and the headline embedding), enforced by a unit test that mutates post-event prices and asserts that no feature changes while the label does. The deployed variant uses context features only, ships off by default behind a configuration gate, and each score is delivered with its additive per-feature contributions and a fixed clause: triage evidence, not a forecast. Once live, every decision is logged and later post-validated against the outcome that actually followed, yielding live precision and calibration monitoring.

Explanation. Each alert is rendered directly from the computed objects (the z-score with its window and threshold, or the retrieved precedents with their similarity scores and measured impact) and ends with an explicit no-prediction disclaimer.

Grounded generation. The components above each answer one question and return one object; nothing in them *combines* those objects, and the joint state space — how many names were flagged, which component dominated each, whether headlines were captured, whether analogous cases exist and which way they went — outgrows any template library. A generative layer performs that synthesis under a construction that prevents the language step from becoming a factual one. Inputs are serialised into an *evidence bundle*: records carrying a citable identifier, a value, and a declared provenance of *measured*, *computed* or *model*. No record carries a *generated* provenance, an invariant asserted by test. The generator sees only the bundle, cites identifiers inline, and its output is verified before delivery; the interface resolves each citation back to its record, so the claim that generated text remains attached to its evidence is a traversal rather than an assertion about a prompt.

Verification applies three conditions. Numbers are bound *per sentence*: the identifiers a sentence cites determine the values it may use, and a value outside that set is a violation even when present elsewhere in the bundle. A global set — the design this replaced — lets a sentence cite one record and quote another's number, which permits a return to be restated as a z-score or a direction to be inverted using the oppositely signed value of a different record. Prediction, advice and asserted causation are rejected by pattern, with the honest disclaimers the system must be able to write removed beforehand by a closed allowlist; detecting negation contextually was tried and abandoned, because a negation placed nearby disables the check. Finally, every sentence making a checkable claim must cite a record. Rejection is per section, and a rejected section is replaced by a deterministic composition that must itself pass the same check.

IV. Evaluation

The evaluation is fixed in advance. The anomaly detector is assessed primarily by the *consistency* of its firing rate across instruments of different volatility (a label-free argument) and secondarily against an extreme-move proxy. The correlation engine is assessed as an information-retrieval task by cross-ticker precision@k against a same-sector relevance proxy, compared with random, recency and lexical baselines over five random seeds. All numbers below are reproduced by versioned scripts.

A. Anomaly detection

Over three years of daily data for fifteen large-cap US tickers, the fixed 3% threshold fires on $\sim 1\%$ of days for a calm stock but $\sim 35\%$ for a volatile one, whereas the z-score fires at a near-constant 2–3% for all. The range of firing rates across tickers is more than twenty times tighter for the z-score (0.015 vs 0.344; Fig. 1). Against the extreme-move proxy the z-score reaches $F_1 = 0.516$ versus 0.218 for the fixed rule, and a window ablation gives F_1 of 0.385, 0.516 and 0.678 for 10-, 20- and 60-day windows.

B. Precedent retrieval

On $3{,}714$ real headlines across five sectors, semantic retrieval clearly outperforms every baseline (Table I). MiniLM reaches a cross-ticker precision@5 of 0.514 (about $2.1\times$ the random base rate), the larger MPNet slightly more (0.538), and both far exceed the lexical baseline (0.346); recency is worst, below the random rate. Small standard deviations

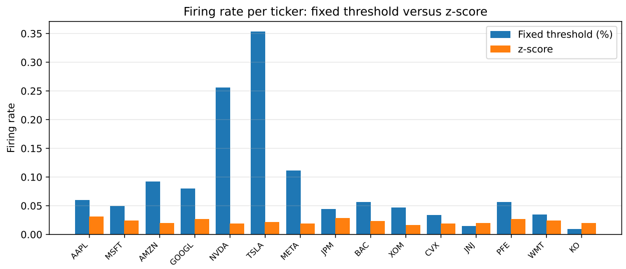

Fig. 1. Firing rate per ticker: a fixed percentage threshold varies dramatically with each stock's volatility, while the volatility-normalised z-score is stable across all tickers.

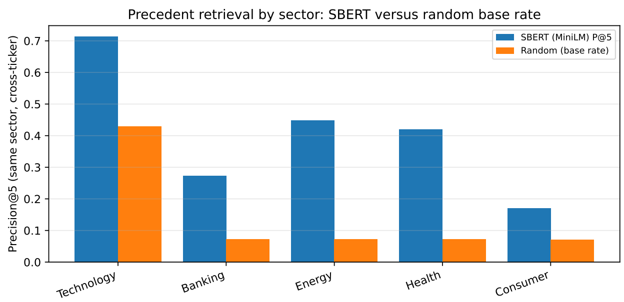

Fig. 2. Per-sector cross-ticker precision@5 against the random base rate. Retrieval beats the base rate in every sector; the lift is largest where the domain vocabulary is most distinctive.

(~ 0.01) show the gap is robust; a follow-up on the multi-year FNSPID corpus ($\sim 80{,}000$ headlines) lifts precision@5 to 0.595, confirming the retrieval at scale. Per sector (Fig. 2), the lift over the base rate is largest where vocabulary is distinctive, in energy ($+0.377$) and health ($+0.348$), and smallest for the broad consumer sector ($+0.100$); technology has the highest raw precision (0.712) only because it dominates the corpus and so has a high base rate (0.429).

C. Faithfulness

Because each alert is rendered from the same objects the system computes, the explanation cannot diverge from the underlying computation; this is checked by an automated test that asserts the rendered alert reproduces each retrieved precedent's date, ticker and measured impact.

For the generative layer, two quantities are measured because they answer different questions. An adversarial corpus of 23 attacks, each of which must be rejected, measures the guard: $23/23$ are blocked. A set of 8 faithful control sentences, each of which must pass, measures whether the guard is merely refusing everything: $8/8$ are admitted. The second is the load-bearing figure, since a guard that rejected all text would score perfectly on the first. Across generated reports, $22/22$ delivered sections satisfy the check and none carrying a violation reaches the reader — a property that must hold by construction and is verified rather than assumed, though it is circular, since the same checker decides and evaluates. What the guard rejects is itself informative: rejections were of asserted causation, advice and prediction rather than of fabricated numbers, indicating that the constraint is doing continuous work. The corpus retains every exploit reproduced against earlier versions of the guard as a permanent regression.

TABLE I
CROSS-TICKER PRECISION@k, MEAN $\pm$ STD OVER FIVE RUNS.

Method	P@5	P@10
SBERT (MiniLM)	0.514 ± 0.015	0.478 ± 0.011
SBERT (MPNet)	$\mathbf{0.538 \pm 0.011}$	$\mathbf{0.506 \pm 0.010}$
Lexical (hashing)	0.346 ± 0.011	0.314 ± 0.008
Recency	0.126 ± 0.006	0.106 ± 0.005
Random (base rate)	0.240 ± 0.004	0.240 ± 0.004

D. Materiality triage

On the multi-year FNSPID corpus [23] (79,753 examples over 2018–2023, fourteen of the fifteen tickers, temporal split by unique days with a five-day embargo; test prevalence 0.378), every trained ranker clears the always-alert floor by a wide margin, but no model that reads the headline text beats the volatility-only baseline: PR-AUC 0.542 for a volatility-only logistic regression, against 0.538 (context-only), 0.496 (context+text), 0.469 (gradient boosting) and 0.439 (text-only), floor 0.378. As a product mechanism, however, triage is clearly worthwhile: within a budget of five alerts per day it raises the share of material alerts from 0.163 to 0.632, with calibrated probabilities (Brier 0.218 against 0.622). The same discipline was applied on the anomaly side, where an Isolation Forest [5] trained causally on identical information lost to the z-score on the identical evaluation region (F_1 0.271 vs 0.530). Both statistical-versus-learned comparisons were pre-committed, and both favoured the transparent choice. A cluster bootstrap and a fair re-test (tuned regularisation, a PCA-reduced text block, a finance-domain encoder) confirm the negative-on-text result is robust rather than an under-fit artefact.

V. DISCUSSION AND LIMITATIONS

The results support the design: transparent volatility-normalised detection is consistent where a fixed threshold is not, semantic retrieval surfaces markedly more sector-analogous precedents than lexical or trivial baselines, and learned triage adds a large budgeted-precision gain while confirming, against its own text features, that the transparent volatility evidence carries the signal. The limitations are stated plainly. Relevance is a same-sector proxy, not a human judgement; the primary retrieval corpus is a recent window from a free service, though a follow-up validated the retrieval at scale on FNSPID (the triage study also used the multi-year corpus); the displayed precedent impact is a raw post-event return, whereas the triage labels are market-adjusted; and no human usefulness study has yet been run, which now covers the generated text as well. The generative layer carries a further limitation stated rather than discovered: its *linguistic* defence is a blocklist, and a blocklist over natural language loses in the limit, which is why the pushed alert path uses a closed-vocabulary allowlist and the two are not claimed to be equivalent. The check confirms that a cited record exists and that a sentence's numbers belong to it, not that the record supports the characterisation in words; and the adversarial exercise behind the current guard was interrupted, with two of six attack perspectives completing and no independent verification stage, so its measured strength is a lower bound. A further, conceptual limitation is that embedding similarity captures *theme* rather than *direction*: a retrieved cluster can mix positive and negative outcomes (a positive headline may retrieve a competitive-threat cluster), so the reported mean is evidence about a theme, not a directional forecast, which is why precedents are always shown individually and every alert disclaims prediction. By design, the system performs no price prediction or trading, so profit-based metrics are out of scope.

VI. Conclusion

InvestiGator shows that useful financial alerting for retail investors can be both transparent and reproducible: a volatility-normalised detector that explains every flag, a news–market correlation engine that retrieves analogous precedents and reports their measured impact as evidence rather than prediction, and a trained, calibrated triage model that nearly quadruples within-budget alert precision while honestly failing to beat a volatility-only baseline with text, and a generative layer whose synthesis stays bound to the records it cites. The last of these carries the most transferable idea: placing generated text beside computed results is safe only if the binding is *per claim* rather than per document, and if the guarantee that binding provides is stated at the strength it actually has. Future work is a market-adjusted impact-magnitude study over the multi-year precedents (retrieval already validated at scale on FNSPID), a human relevance and usefulness study covering the generated reports, completing the interrupted adversarial evaluation of the guard, and closing the triage text gap with richer text than headlines.

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